

Development and Explanation

Frontend Development

Overview

The frontend of the **DarshanEase** application is developed using **React.js**, **TypeScript**, **Vite**, and **Tailwind CSS**. It provides a responsive, interactive, and user-friendly interface that enables devotees to book darshan tickets online and allows administrators to manage temples, bookings, and users. The frontend communicates with the backend through RESTful APIs using Axios.

Home Page

The Home Page is the landing page of the application. It introduces users to the DarshanEase platform and provides quick navigation to major features.

Features

- Responsive navigation bar
- Hero section with welcome message
- Featured temples
- About DarshanEase section
- Footer with contact information

Purpose

The Home Page creates the first impression of the application and guides users to register, log in, or browse temples.

User Registration

The Registration Page allows new devotees to create an account.

Fields

- Full Name
- Email Address
- Phone Number
- Password

- Confirm Password

Process

1. User enters the required details.
2. Client-side validation checks the input.
3. Data is sent to the backend using an API request.
4. The backend stores the user information in MongoDB.
5. A success message is displayed.

User Login

The Login Page authenticates registered users.

Features

- Email and password authentication
- JWT-based login
- Error handling for invalid credentials

Workflow

1. User enters login credentials.
2. Frontend sends a login request.
3. Backend verifies the credentials.
4. JWT token is returned.
5. Token is stored securely.
6. User is redirected to the dashboard.

Temple Listing Page

This page displays all temples available in the system.

Features

- Temple cards

- Temple image
- Temple name
- Location
- View Details button

Purpose

Allows users to browse and select a temple for darshan booking.

Temple Details Page

Displays detailed information about the selected temple.

Information Displayed

- Temple name
- Description
- Address
- Opening and closing timings
- Available darshan slots

Purpose

Helps users understand temple information before booking.

Darshan Slot Selection

Users can view and select available darshan slots.

Features

- Date selection
- Time slot selection
- Seat availability
- Book Now button

Workflow

1. User selects a preferred slot.
2. Frontend validates the selection.
3. Selected slot details are forwarded to the booking page.

Ticket Booking Page

Allows users to confirm their booking.

Features

- Booking summary
- Number of devotees
- Booking confirmation

Workflow

1. User reviews booking details.
2. Clicks **Confirm Booking**.
3. Booking request is sent to the backend.
4. Booking confirmation is displayed.

Booking History Page

Displays all bookings made by the logged-in user.

Features

- Booking ID
- Temple Name
- Date and Time
- Booking Status
- Cancel Booking option

Purpose

Allows users to manage and review their reservations.

User Profile

Displays user account information.

Features

- View profile
- Update profile information
- Change password
- Logout

Admin Dashboard

The Admin Dashboard provides complete administrative control over the application.

Modules

Temple Management

- Add Temple
- Edit Temple
- Delete Temple

Slot Management

- Create Slot
- Update Slot
- Delete Slot

Booking Management

- View Bookings
- Cancel Bookings
- Update Booking Status

User Management

- View Users

- Manage User Accounts

Reports

- Total Users
- Total Bookings
- Available Slots
- Booking Statistics

API Integration

Axios is used for communication between the frontend and backend.

API Services

- Authentication Service
- Temple Service
- Booking Service
- Slot Service
- Admin Service

Each service sends HTTP requests and receives JSON responses.

State Management

React Context API is used for global state management.

Managed Data

- Logged-in user
- Authentication status
- JWT token
- User profile

Routing

React Router provides client-side navigation.

Routes

- /
- /login
- /register
- /temples
- /temples/:id
- /book
- /history
- /profile
- /admin

Protected routes ensure that only authenticated users can access restricted pages.

Responsive Design

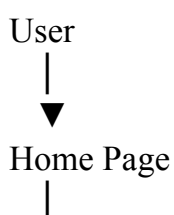
Tailwind CSS is used to build a responsive interface.

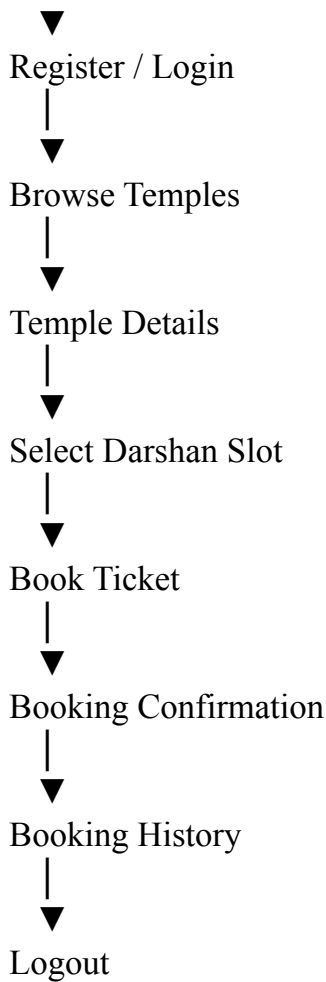
Supported Devices

- Desktop
- Laptop
- Tablet
- Mobile

Responsive layouts automatically adapt to different screen sizes.

Frontend Workflow





Technologies Used

Technology	Purpose
React.js	Build user interface
TypeScript	Type safety and maintainability
Vite	Fast development and build tool
Tailwind CSS	Responsive styling
React Router	Client-side routing
Axios	REST API communication
React Context API	Global state management

Advantages of the Frontend

- Responsive user interface.
- Modular component-based architecture.

- Fast rendering with React and Vite.
- Secure authentication using JWT.
- Easy navigation with React Router.
- Clean and maintainable code structure.
- Scalable for future enhancements such as online payments, notifications, QR code support, and multilingual interfaces.

Conclusion

The frontend of DarshanEase is designed to provide a seamless and intuitive experience for devotees and administrators. By leveraging React.js, TypeScript, Tailwind CSS, and RESTful API integration, the application delivers efficient navigation, secure authentication, responsive layouts, and smooth interaction with the backend. The modular architecture ensures easy maintenance, scalability, and support for future feature enhancements.